Lab 7: Fuzz Testing with Hypothesis & Pytest

Name: Rohil Thimmanahalli Girish | SRN: PES1UG24CS384 | Section: G

GitHub Repo: https://github.com/rohilgirish/SE-Labs-PES1UG24CS384

1. Execution of the Original Application (food_delivery.py)

```
PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7> python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:2,burger:1
Order: {'pizza': 2, 'burger': 1}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 4.5
ETA in Minutes: 33

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7> python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza
Traceback (most recent call last):
  File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 31, in <module>
    main()
  File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 17, in main
    result = parse_order(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 4, in parse_order
    name, qty = item.split(':')
    ^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

2. Failing Test Cases Identified During Fuzz Testing

```
PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7> python -m pytest test_food_delivery.py -v
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-
7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7
plugins: hypothesis-6.170.15
collecting ... collected 7 items

test_food_delivery.py::test_parse_order_no_crash FAILED                  [ 14%]
test_food_delivery.py::test_delivery_eta_no_crash FAILED                 [ 28%]
test_food_delivery.py::test_coupon_discount_no_crash FAILED              [ 42%]
test_food_delivery.py::test_non_string_inputs_no_crash FAILED            [ 57%]
test_food_delivery.py::test_delivery_eta_valid_distance PASSED           [ 71%]
test_food_delivery.py::test_coupon_discount_valid_code PASSED            [ 85%]
test_food_delivery.py::test_parse_order_valid_csv FAILED                 [100%]

================================== FAILURES ===================================
_____________________________ test_parse_order_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 7, in test_pars
e_order_no_crash
  |     def test_parse_order_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 9, in test_pa
rse_order_no_crash
    |     result = parse_order(s)
    |              ^^^^^^^^^^^^^^
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 5, in parse_order
    |     order[name.strip()] = int(qty)
    |                           ^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 11, in test_p
arse_order_no_crash
    |     assert False, f"parse_order crashed with: {e}"
    | AssertionError: parse_order crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 9, in test_pa
rse_order_no_crash
    |     result = parse_order(s)
    |              ^^^^^^^^^^^^^^
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 4, in parse_order
    |     name, qty = item.split(':')
    |     ^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 11, in test_p
arse_order_no_crash
```

2. Failing Test Cases Identified During Fuzz Testing (Part 2/5)

```
    |     assert False, f"parse_order crashed with: {e}"
    | AssertionError: parse_order crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 9, in test_pa
rse_order_no_crash
    |     result = parse_order(s)
    |              ^^^^^^^^^^^^^^
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 3, in parse_order
    |     for item in items_csv.split(','):
    |                 ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 11, in test_p
arse_order_no_crash
    |     assert False, f"parse_order crashed with: {e}"
    | AssertionError: parse_order crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_delivery_eta_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 19, in test_del
ivery_eta_no_crash
  |     def test_delivery_eta_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 21, in test_d
elivery_eta_no_crash
    |     result = delivery_eta(s)
    |              ^^^^^^^^^^^^^^
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 9, in delivery_eta
    |     return int(15 + float(distance_km) * 4)
    |                     ^^^^^^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 23, in test_d
elivery_eta_no_crash
    |     assert False, f"delivery_eta crashed with: {e}"
    | AssertionError: delivery_eta crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_delivery_eta_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 21, in test_d
elivery_eta_no_crash
    |     result = delivery_eta(s)
    |              ^^^^^^^^^^^^^^
```

2. Failing Test Cases Identified During Fuzz Testing (Part 3/5)

```
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 9, in delivery_eta
    |     return int(15 + float(distance_km) * 4)
    |                    ^^^^^^^^^^^^^^^^^^
    | TypeError: float() argument must be a string or a real number, not 'NoneType'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 23, in test_d
elivery_eta_no_crash
    |     assert False, f"delivery_eta crashed with: {e}"
    | AssertionError: delivery_eta crashed with: float() argument must be a string or a real number, not 'No
neType'
    | assert False
    | Failing test case: test_delivery_eta_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_coupon_discount_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 28, in test_cou
pon_discount_no_crash
  |     def test_coupon_discount_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 30, in test_c
oupon_discount_no_crash
    |     result = coupon_discount(s)
    |              ^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 12, in coupon_disc
ount
    |     return int(code.upper().split('SAVE')[1])
    |                ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 32, in test_c
oupon_discount_no_crash
    |     assert False, f"coupon_discount crashed with: {e}"
    | AssertionError: coupon_discount crashed with: list index out of range
    | assert False
    | Failing test case: test_coupon_discount_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 30, in test_c
oupon_discount_no_crash
    |     result = coupon_discount(s)
    |              ^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\food_delivery.py", line 12, in coupon_disc
ount
    |     return int(code.upper().split('SAVE')[1])
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'upper'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7\test_food_delivery.py", line 32, in test_c
oupon_discount_no_crash
    |     assert False, f"coupon_discount crashed with: {e}"
    | AssertionError: coupon_discount crashed with: 'NoneType' object has no attribute 'upper'
```

2. Failing Test Cases Identified During Fuzz Testing (Part 4/5)

```
    | assert False
    | Failing test case: test_coupon_discount_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_non_string_inputs_no_crash _____________________

val = 0

    @given(st.integers() | st.floats() | st.booleans() | st.lists(st.text()))
    def test_non_string_inputs_no_crash(val):
        try:
>           assert isinstance(parse_order(val), dict)
                              ^^^^^^^^^^^^^^^^

test_food_delivery.py:39:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

items_csv = 0

    def parse_order(items_csv):
        order = {}
>       for item in items_csv.split(','):
                    ^^^^^^^^^^^^^^^^
E       AttributeError: 'int' object has no attribute 'split'

food_delivery.py:3: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.integers() | st.floats() | st.booleans() | st.lists(st.text()))
>   def test_non_string_inputs_no_crash(val):
                ^^^

test_food_delivery.py:37:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

val = 0

    @given(st.integers() | st.floats() | st.booleans() | st.lists(st.text()))
    def test_non_string_inputs_no_crash(val):
        try:
            assert isinstance(parse_order(val), dict)
            eta = delivery_eta(val)
            assert eta is None or (isinstance(eta, int) and eta >= 15)
            discount = coupon_discount(val)
            assert isinstance(discount, int) and discount >= 0
        except Exception as e:
>           assert False, f"crashed on non-string input {val!r}: {e}"
E           AssertionError: crashed on non-string input 0: 'int' object has no attribute 'split'
E           assert False
E           Failing test case: test_non_string_inputs_no_crash(
E               val=0,  # or any other generated value
E           )

test_food_delivery.py:45: AssertionError
___________________________ test_parse_order_valid_csv ____________________________
```

2. Failing Test Cases Identified During Fuzz Testing (Part 5/5)

```
    def test_parse_order_valid_csv():
        assert parse_order("pizza:2,burger:1") == {"pizza": 2, "burger": 1}
>       assert parse_order(" pizza : 2 , pizza : 1 ") == {"pizza": 3}
E       AssertionError: assert {'pizza': 1} == {'pizza': 3}
E
E         Differing items:
E         {'pizza': 1} != {'pizza': 3}
E
E         Full diff: (-: missing in left side, +: extra in left side)
E           {
E         -     'pizza': 3,...
E
E         ...Full output truncated, use '-vv' to show

test_food_delivery.py:62: AssertionError
========================== short test summary info ==========================
FAILED test_food_delivery.py::test_parse_order_no_crash - ExceptionGroup: Hyp...
FAILED test_food_delivery.py::test_delivery_eta_no_crash - ExceptionGroup: Hy...
FAILED test_food_delivery.py::test_coupon_discount_no_crash - ExceptionGroup:...
FAILED test_food_delivery.py::test_non_string_inputs_no_crash - AssertionErro...
FAILED test_food_delivery.py::test_parse_order_valid_csv - AssertionError: as...
======================== 5 failed, 2 passed in 5.53s ========================
```

3. Execution of the Corrected Application (food_delivery.py)

```
PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7> python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:2,burger:1
Order: {'pizza': 2, 'burger': 1}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 4.5
ETA in Minutes: 33

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50

PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7> python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza
Order: {}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): abc
ETA in Minutes: None

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): INVALID
Discount: 0
```

4. Final Test Execution Showing Successful Results

```
PS C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7> python -m pytest test_food_delivery.py -v
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-
7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Lenovo\Desktop\Sem-5\SE\SE-lab\Lab-7\Lab 7
plugins: hypothesis-6.170.15
collecting ... collected 7 items

test_food_delivery.py::test_parse_order_no_crash PASSED                  [ 14%]
test_food_delivery.py::test_delivery_eta_no_crash PASSED                 [ 28%]
test_food_delivery.py::test_coupon_discount_no_crash PASSED              [ 42%]
test_food_delivery.py::test_non_string_inputs_no_crash PASSED            [ 57%]
test_food_delivery.py::test_delivery_eta_valid_distance PASSED           [ 71%]
test_food_delivery.py::test_coupon_discount_valid_code PASSED            [ 85%]
test_food_delivery.py::test_parse_order_valid_csv PASSED                 [100%]

============================== 7 passed in 1.58s ==============================
```

5. Fixed Python Source Code: food_delivery.py

```
import math
import re


def parse_order(items_csv):
    """Parse a 'name:qty,...' CSV string into a dict of positive item quantities."""
    if not isinstance(items_csv, str) or not items_csv.strip():
        return {}

    order = {}
    for raw_item in items_csv.split(','):
        item = raw_item.strip()
        if not item:
            continue
        parts = item.split(':')
        if len(parts) != 2:
            continue
        name = parts[0].strip()
        qty_str = parts[1].strip()
        if not name or not qty_str.isascii() or not qty_str.isdigit() or len(qty_str) > 9:
            continue
        qty = int(qty_str)
        if qty <= 0:
            continue
        order[name] = order.get(name, 0) + qty
    return order


def delivery_eta(distance_km):
    """Calculate delivery ETA in minutes from a non-negative finite distance in km."""
    if isinstance(distance_km, bool) or not isinstance(distance_km, (int, float, str)):
        return None
    if isinstance(distance_km, str):
        distance_km = distance_km.strip()
        if not distance_km:
            return None
    try:
        dist = float(distance_km)
    except (ValueError, TypeError, OverflowError):
        return None
    if not math.isfinite(dist) or dist < 0 or dist > 10000:
        return None
    eta = 15 + dist * 4
    if not math.isfinite(eta):
        return None
    return int(eta)


def coupon_discount(code):
    """Extract a non-negative integer discount from a SAVE<amount> coupon code."""
    if not isinstance(code, str):
        return 0
    cleaned = code.strip().upper()
    match = re.fullmatch(r"SAVE([0-9]+)", cleaned)
    if not match:
        return 0
    digits = match.group(1)
    if len(digits) > 6:
        return 0
    return int(digits)


def main():
    print("==== Running parse_order ====")
    user_input = input("Enter items as name:qty CSV (e.g. pizza:2,burger:1): ")
    result = parse_order(user_input)
    print("Order:", result)

    print("\n==== Running delivery_eta ====")
```

```
    user_input = input("Enter distance in km (e.g. 4.5): ")
    result = delivery_eta(user_input)
    print("ETA in Minutes:", result)

    print("\n==== Running coupon_discount ====")
    user_input = input("Enter a coupon code (e.g. SAVE50): ")
    result = coupon_discount(user_input)
    print("Discount:", result)


if __name__ == "__main__":
    main()
```

6. Completed Fuzz Test Source Code: test_food_delivery.py

```
import math
from hypothesis import given, strategies as st
from food_delivery import parse_order, delivery_eta, coupon_discount


@given(st.text() | st.none())
def test_parse_order_no_crash(s):
    try:
        result = parse_order(s)
    except Exception as e:
        assert False, f"parse_order crashed with: {e}"
    assert isinstance(result, dict)
    for k, v in result.items():
        assert isinstance(k, str) and len(k) > 0
        assert isinstance(v, int) and v > 0


@given(st.text() | st.none())
def test_delivery_eta_no_crash(s):
    try:
        result = delivery_eta(s)
    except Exception as e:
        assert False, f"delivery_eta crashed with: {e}"
    assert result is None or (isinstance(result, int) and not isinstance(result, bool) and result >= 15)


@given(st.text() | st.none())
def test_coupon_discount_no_crash(s):
    try:
        result = coupon_discount(s)
    except Exception as e:
        assert False, f"coupon_discount crashed with: {e}"
    assert isinstance(result, int) and not isinstance(result, bool) and result >= 0


@given(st.integers() | st.floats() | st.booleans() | st.lists(st.text()))
def test_non_string_inputs_no_crash(val):
    try:
        assert isinstance(parse_order(val), dict)
        eta = delivery_eta(val)
        assert eta is None or (isinstance(eta, int) and eta >= 15)
        discount = coupon_discount(val)
        assert isinstance(discount, int) and discount >= 0
    except Exception as e:
        assert False, f"crashed on non-string input {val!r}: {e}"


@given(st.floats(min_value=0, max_value=1000, allow_nan=False, allow_infinity=False))
def test_delivery_eta_valid_distance(d):
    result = delivery_eta(str(d))
    assert result == int(15 + d * 4)


@given(st.integers(min_value=0, max_value=10000))
def test_coupon_discount_valid_code(amount):
    assert coupon_discount(f"save{amount}") == amount
    assert coupon_discount(f"  SAVE{amount}  ") == amount


def test_parse_order_valid_csv():
    assert parse_order("pizza:2,burger:1") == {"pizza": 2, "burger": 1}
    assert parse_order(" pizza : 2 , pizza : 1 ") == {"pizza": 3}
```